



BSLD (Beacon Signal Location Device)

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Pisitpong
Vorachai
Tappaya**

**Jaraspornjaroen
Wannathanom
Chananithitham
Kaewbut**

**BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY 2022

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A SENIOR PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

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THIS SENIOR PROJECT HAS BEEN APPROVED
TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING 2022

EXAMINING COMMITTEE

..... ADVISOR
(Asst. Prof. Dr.Suppakarn Chansareewittaya)

.....COMMITTEE
(Aj. Dr.Mahamah Sebakor)

.....COMMITTEE
(Aj. Ronnachai Sretawat Na Ayutaya)

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Lastly, we would like to thanks any committee and any suggestion from everyone from helping us improving the quality of our work and made this project in a best state possible.

Phuektinan	Jaraspornjaroen
Pisitpong	Wannathanom
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Title	BSLD (Beacon Signal Location Device)	
Author	Phuektinan	Jaraspornjaroen
	Pisitpong	Wannathanom
	Vorachai	Chananithitham
	Tappaya	Kaewbut
Degree	Bachelor of Engineering	
Supervisory Committee	Asst. Prof. Dr.Suppakarn Chansareewittaya	Advisor
	Aj. Dr.Mahamah Sebakor	Committee
	Aj. Ronnachai Sretawat Na Ayutaya	Committee

Abstract

BSLD Project is designed to help those interested in testing distance measurement with Wi-Fi signals to be able to measure in many areas, such as open areas, areas with obstacles. By finding the intersection point of the area, using the ESP32 device as a WiFi signal transmitter. And the test has been used in many places to compare with the use of manual measuring equipment such as tape measure and ruler that are taken down to measure in real areas.

that we have made this project We sincerely hope that this experiment will be of benefit to those interested in our experiment. They are also interested in taking it for further study. and taking our project for further development in the future

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CHAPTER 1

INTRODUCTION

1.1 Background and Rational

At present, measuring space with manual measuring devices may not always be accurate. Sometimes, areas with obstacles may cause the distance measured to be different from the actual distance of that area. and measuring space in open areas with no obstructions using hand-operated measuring equipment. It is not always accurate as well because there may be discrepancies in the distance from the distance measurer changing the measuring position. or the distance is inaccurate because the measurer forgets the distance of the area being measured Therefore, we have invented a device that helps in measuring distances in various areas. to be compared with those measured by hand

1.2 Objective

1.2.1 Build a machine that can measure distance successfully

1.2.2 Experiment with measuring the actual distance and measuring the distance from the machine to see how much discrepancy there is.

1.3 Scope

1.3.1 Experiment with an open, unobstructed area of 5x5.

1.3.2 Experiment with a location with 5x5 obstacles.

1.4 Methodology

1.4.1 Proposal preparation and defense

1.4.2 Literature review and requirement gathering

1.4.3 System analysis

1.4.4 System design

1.4.5 Senior Project 1 document preparation

1.4.6 Senior Project 1 exam

1.4.7 System development

1.4.8 System testing

1.4.9 Senior Project 2 exam

1.4.10 Senior Project 2 exam

1.6 Expected Result

1.6.1 BSLD was developed.

1.6.2 Can be used to measure distance and compare with actual distance.

1.7 Place and Equipment

1.7.1 Place

1.7.1.1 In front of building S7

1.7.1.2 Building E4 parking lot

1.7.2 Hardware

1.7.2.1 Esp32

1.7.3 Software

1.7.3.1 Arduino IDE

CHAPTER 2

LITERATURE REVIEW

2.1 Related Research

1.Didi Surian a, Vitaliy Kim a, Ranjeeta Menon b c, Adam G. Dunn a, Vitali Sintchenko b c, Enrico Coiera a a.) Centre for Health Informatics, Australian Institute of Health Innovation, Macquarie University, Sydney, Australia b.) Sydney Medical School and Marie Bashir Institute for Infectious Diseases and Biosecurity, The University of Sydney, Sydney, Australia c.) Centre for Infectious Disease and Microbiology-Public Health, ICPMR, Westmead Hospital, Sydney, Australia has research about tracking the locations of individuals in indoor environments. We can study from this research to apply to our project.

[Website <https://www.sciencedirect.com/science/article/pii/S1532046419302072>]

2.Hao Zhu a, Yaohua Xie b a.)Information Center University of Electronic Science and Technology of China Chengdu, China b.)Yale University New Haven, U.S.A has research on application of beacon-based Location Technology in America Universities.

[Website <https://iopscience.iop.org/article/10.1088/1755-1315/252/5/052002/pdf>]

2.2 Related Work

1. Minew BLE beacon is a bluetooth low energy beacon that enables real-time indoor positioning and tracking of people or assets and also we can locate them in real time.

2.RTLS Technology by Blueiots provide both hardware and software which is combined together to track indoor assets or people.

2.3 Related Technology

1. **ESP32** is a low-cost, low-power Microcontroller with integrated Wi-Fi and Bluetooth. It is the successor to the ESP8266 which is also a low-cost Wi-Fi microchip albeit with limited vastly limited functionality. It is an integrated antenna and RF balun, power amplifier, low-noise amplifiers, filters, and power management module. The entire solution takes up the least amount of printed circuit board area. This board is used with 2.4 GHz dual-mode Wi-Fi and Bluetooth chips by TSMC 40nm low power technology, power and RF properties best, which is safe, reliable, and scale-able to a variety of applications.



figure 2.4(1) Esp32

2.3.1 Software

1. **Arduino Software (IDE)** - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them.

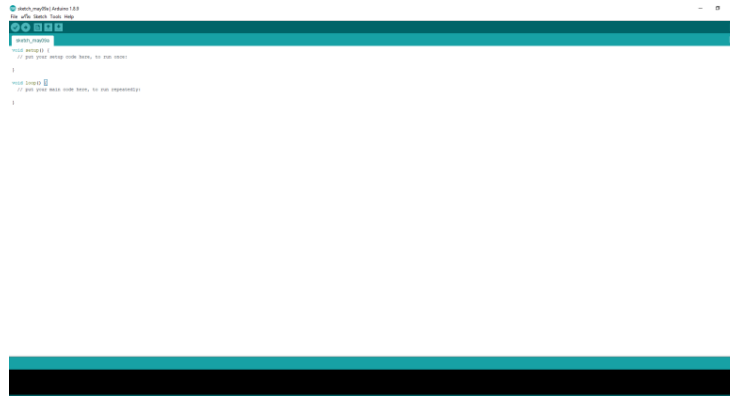


figure 2.4.1 Arduino Software

2.3.2 Language and tool development

1. **C** is a general-purpose programming language created by Dennis Ritchie at the Bell Laboratories in 1972. It is a very popular language, despite being old. C is strongly associated with UNIX, as it was developed to write the UNIX operating system.

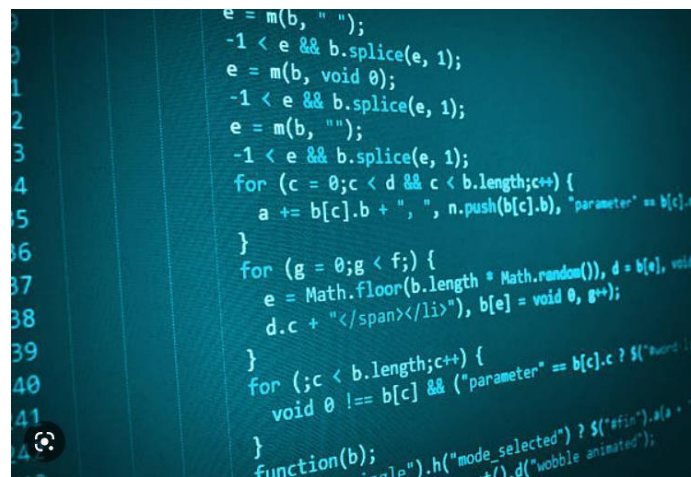


figure 2.4.2 C Language

CHAPTER 3

ANALYSIS AND DESIGN

3.1 Existing System

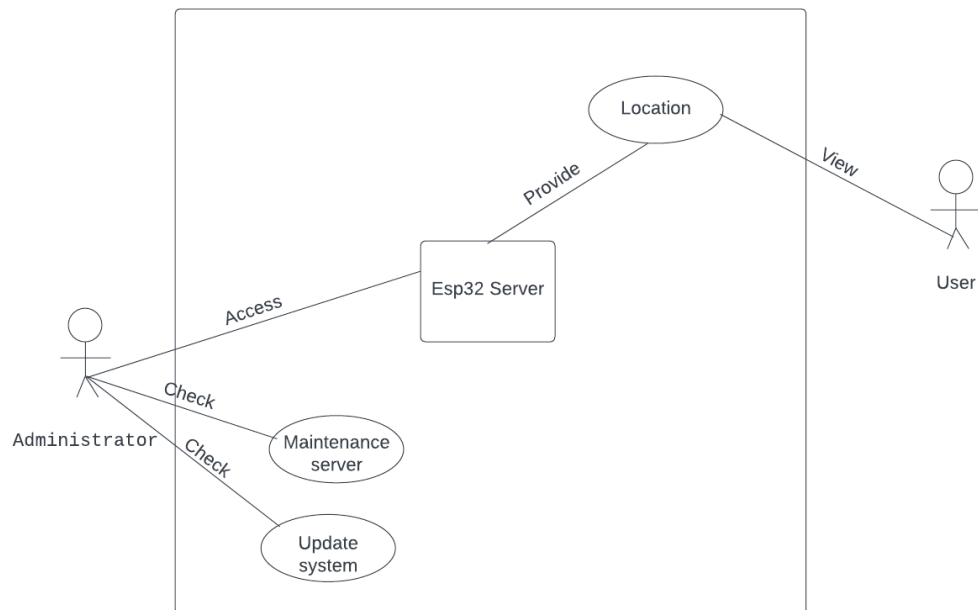


figure 3.1 Use case diagram of the existing system

It shows the skeleton system of BSLD, divided into admin and user side, with Esp32 intermediary. Server is a central control system.

3.2 New System

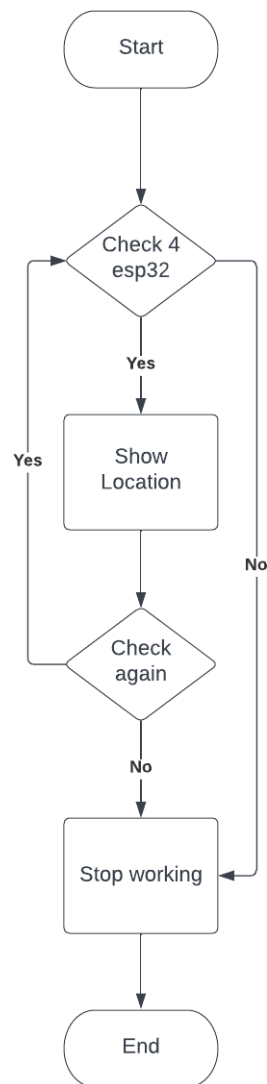


figure 3.2 Use case diagram of the new system

It works by turning on the machine and wiping whether Esp is complete in all 4 corners of the room or not. When 4 corners are complete, the system will measure coordinates to identify the location for the user to see. When finished viewing, the user can turn off the machine. usability

3.3 Use case diagram

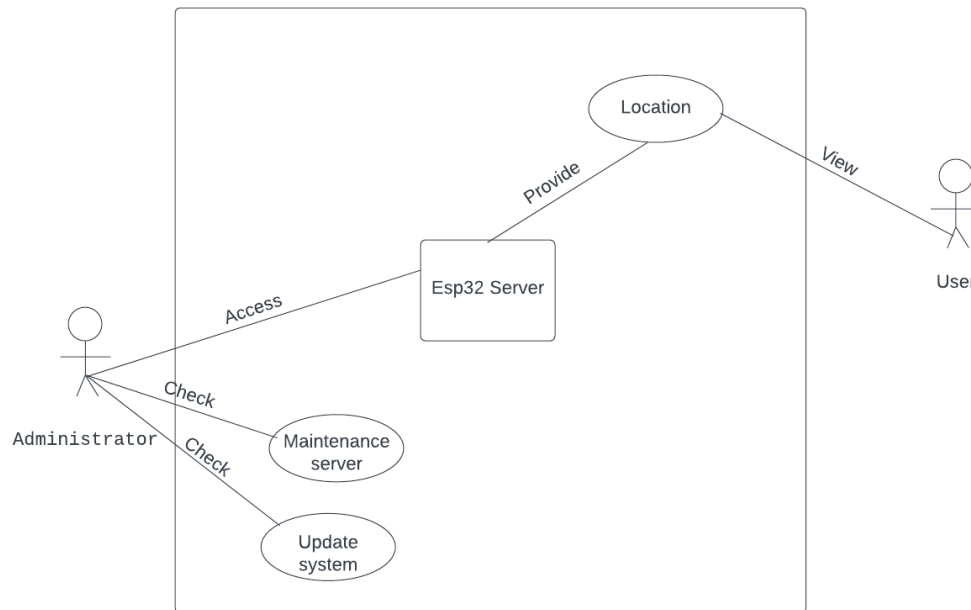


figure 3.3 Use case diagram of the new system

We have Esp32 to control the system by working that will allow the user to see the location and if there is an error from the power-on process and check if there is Esp32. complete or not, the administrator can come in to fix and repair such errors

3.4 Activity diagram

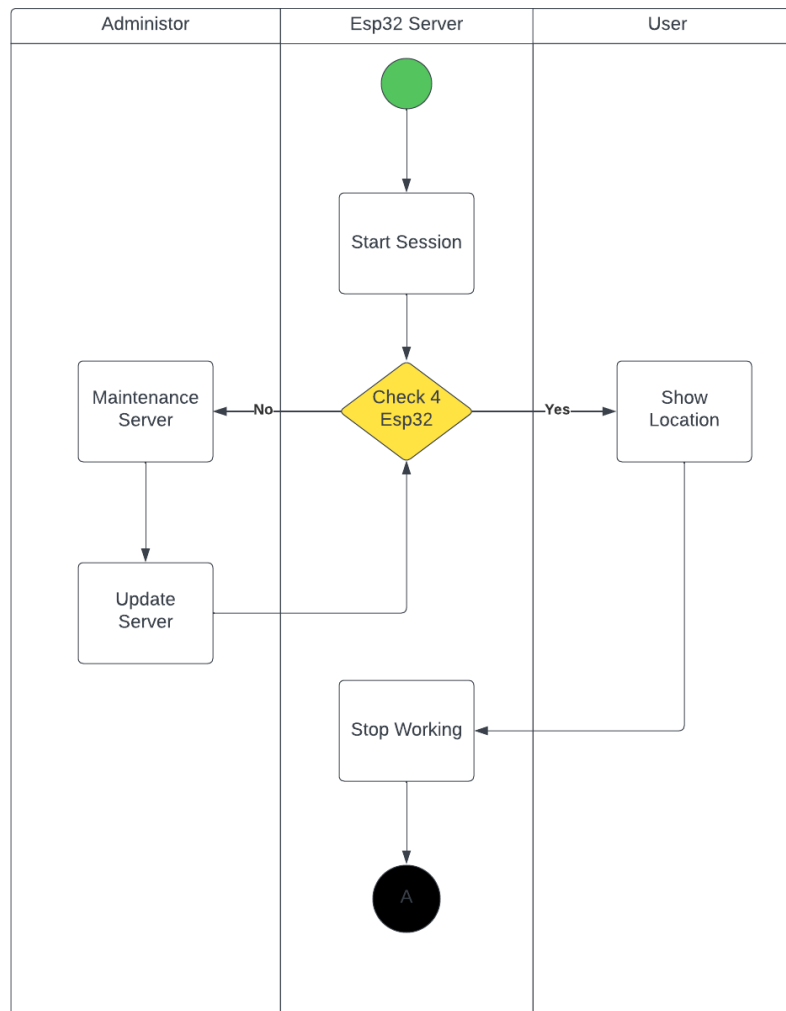


figure 3.4 Activity diagram of the new system

From the figure, there are 3 systems that can be divided into admin esp32 user. The main duty of admin is to take care and repair errors of Esp32 where Esp32 will show the location to the user. esp32 all 4 or not, if not complete, will inform the admin to take care of it in case you can speak and use it can see the location as usual

CHAPTER 4

RESULTS

In this project, we built a machine to measure distance using a Wi-Fi signal using ESP32 and tested it with real distance measurement to see the discrepancy.

4.1 Device BSLD

Our machine displays the intersection of the x and y axes via a serial monitor.



figure 4.1 Device BSLD

4.2 Experiment

4.2.1 In the experiment, we measured in the open air using a machine to measure the actual distance.



figure 4.2.1 measured in the open air using a machine to measure the actual distance.

4.2.2 We approached the corner about 1 meter from the edge and stored the values obtained from the machine.



figure 4.2.2 Approached the corner about 1 meter from the edge and stored the values obtained from the machine

2.3 Change location to Building E4 parking lot, which is a place with obstacles.



figure 4.2.3 Change location to Building E4 parking lot, which is a place with obstacles.

4.2.4 Collect the values received through the serial monitor.

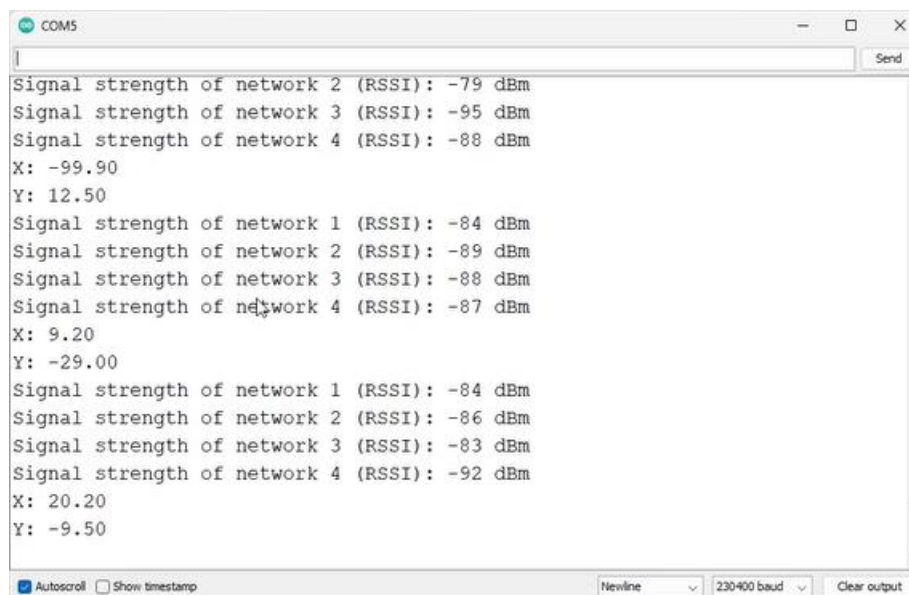


figure 4.2.4 Collect the values received through the serial monitor.

4.3 Experiment Result

4.3.1 Results of the experiment At the open area In front of building S7

	RSSI value from machine 1	RSSI value from machine 2	RSSI value from machine 3	RSSI value from machine 4	x and y axis values
Approaching corner 1	-60 dBm	-77 dBm	-79 dBm	-91 dBm	X = -63.8 Y = -31
Approaching corner 2	-73 dBm	-61 dBm	-78 dBm	-82 dBm	X = -46.8 Y = 31.3
Approaching corner 3	-82 dBm	-78 dBm	-58 dBm	-64 dBm	X = 74.5 Y = 2.5
Approaching corner 4	-80 dBm	-86 dBm	-75 dBm	-60 dBm	X = 58.6 Y = -31.1
At the center	-72 dBm	-73 dBm	-72 dBm	-71 dBm	X = -1 Y = 6

4.3.2 Results of the experiment At where obstacles are Building E4 parking lot

	RSSI value from machine 1	RSSI value from machine 2	RSSI value from machine 3	RSSI value from machine 4	x and y axis values
Approaching corner 1	-65 dBm	-74 dBm	-77 dBm	Lost Connect	-
Approaching corner 2	-69 dBm	-66 dBm	Lost Connect	-72 dBm	-
Approaching corner 3	-84 dBm	Lost Connect	-67 dBm	-88 dBm	-
Approaching corner 4	Lost Connect	-83 dBm	-74 dBm	-57 dBm	-
At the center	-78 dBm	-79 dBm	-75 dBm	-70 dBm	X = 20 Y = -2.2

CHAPTER 5

CONCLUSION

5.1 Conclusion

The BSLD project is an experiment in using wi-fi signals to measure distance to compare the expected deviation from the actual distance. The working principle is to use ESP32 to emit signals and receive signals to store values and reveal defects and problems that need to be solved. The fever of using wi-fi signals and causing those who are interested in this system to use it for further development from the experiment.

5.2 Discussion

In doing this project, there were many problems because the use of ESP32 to transmit the signal caused many limitations, such as the distance of the signal being very close. When the distance was too large, the signal was lost and The distance measured by the machine had too much deviation from the actual distance measured. Many variables cause high deviations, such as obscuring objects and noise and noise from various places. If this occurs having a larger budget for the project might allow the use of a quality signal transmitter and make the problems encountered disappear.

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APPENDIX

SOURCE CODE

First, an example of Arduino files.

```

#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <WiFi.h>

const char* ssid1 = "MyESP32AP1";
const char* password1 = "1234567890";
const char* ssid2 = "MyESP32AP2";
const char* password2 = "1234567890";
const char* ssid3 = "MyESP32AP3";
const char* password3 = "1234567890";
const char* ssid4 = "MyESP32AP4";
const char* password4 = "1234567890";

unsigned long lastReadTime = 0;
unsigned long interval = 5000;

int measured_rssi = -55;

int signal_strength1;
int signal_strength2;
int signal_strength3;
int signal_strength4;

struct Point {
    float x;
    float y;
};

// OLED display configuration
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
#define SSD1306_I2C_ADDRESS 0x3C
#define SSD1306_I2C_RESET -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

// Function declarations
float calculateDistance(int measured_rssi, int reference_rssi);
Point calculateCoordinates(int x1, int y1, int x2, int y2, int x3, int y3, int x4, int y4, int measured_rssi,
int signal_strength1, int signal_strength2, int signal_strength3, int signal_strength4);

void setup() {
    Serial.begin(230400);
    delay(10);

```

```

// Initialize the OLED display
if (!display.begin(SSD1306_I2C_ADDRESS, 0x3C, SSD1306_I2C_RESET)) {
    Serial.println(F("SSD1306 allocation failed"));
    for (;;)
}
delay(2000);
display.clearDisplay();
display.display();

WiFi.begin(ssid1, password1);
Serial.println();
Serial.print("Connecting to ");
Serial.println(ssid1);

while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}

Serial.println("");
Serial.println("WiFi 1 connected");
}

void loop() {
    if (millis() - lastReadTime >= interval) {
        lastReadTime = millis();

        WiFi.disconnect();
        WiFi.begin(ssid1, password1);
        while (WiFi.status() != WL_CONNECTED) {
            delay(500);
        }
        signal_strength1 = WiFi.RSSI();
        Serial.print("Signal strength of network 1 (RSSI): ");
        Serial.print(signal_strength1);
        Serial.println(" dBm");

        WiFi.disconnect();
        WiFi.begin(ssid2, password2);
        while (WiFi.status() != WL_CONNECTED) {
            delay(500);
        }
        signal_strength2 = WiFi.RSSI();
        Serial.print("Signal strength of network 2 (RSSI): ");
        Serial.print(signal_strength2);
        Serial.println(" dBm");

        WiFi.disconnect();
        WiFi.begin(ssid3, password3);
        while (WiFi.status() != WL_CONNECTED) {
            delay(500);
        }
        signal_strength3 = WiFi.RSSI();
        Serial.print("Signal strength of network 3 (RSSI): ");
        Serial.print(signal_strength3);
        Serial.println(" dBm");
    }
}

```

```

WiFi.disconnect();
WiFi.begin(ssid4, password4);
while (WiFi.status() != WL_CONNECTED) {
    delay(500);
}
signal_strength4 = WiFi.RSSI();
Serial.print("Signal strength of network 4 (RSSI): ");
Serial.print(signal_strength4);
Serial.println(" dBm");

int x1 = 0;
int y1 = 0;
int x2 = 0; // Room size
int y2 = 10;
int x3 = 10;
int y3 = 10;
int x4 = 10;
int y4 = 0;

Point estimatedCoordinates = calculateCoordinates(x1, y1, x2, y2, x3, y3, x4, y4, measured_rssi,
signal_strength1, signal_strength2, signal_strength3, signal_strength4);

// Print the estimated X and Y coordinates
Serial.print("X: ");
Serial.println(estimatedCoordinates.x);
Serial.print("Y: ");
Serial.println(estimatedCoordinates.y);

// Display coordinates on the OLED display
display.clearDisplay();
display.setTextSize(1);
display.setTextColor(SSD1306_WHITE);
display.setCursor(0, 0);
display.print("X: ");
display.print(estimatedCoordinates.x, 2); // Display X with two decimal places
display.println();
display.print("Y: ");
display.print(estimatedCoordinates.y, 2); // Display Y with two decimal places
display.display();
}
}

float calculateDistance(int measured_rssi, int reference_rssi) {
    float distance = measured_rssi - reference_rssi;
    return distance;
}

Point calculateCoordinates(int x1, int y1, int x2, int y2, int x3, int y3, int x4, int y4, int measured_rssi,
int signal_strength1, int signal_strength2, int signal_strength3, int signal_strength4) {
    Point estimatedPoint;

    // Calculate distances from measured RSSI
    float distance1 = calculateDistance(measured_rssi, signal_strength1);
    float distance2 = calculateDistance(measured_rssi, signal_strength2);
    float distance3 = calculateDistance(measured_rssi, signal_strength3);

```

```

float distance4 = calculateDistance(measured_rssi, signal_strength4);

float A1 = 2 * (x2 - x1);
float B1 = 2 * (y2 - y1);
float C1 = (distance1 * distance1 - distance2 * distance2) - (x1 * x1 - x2 * x2) - (y1 * y1 - y2 * y2);

float A2 = 2 * (x3 - x1);
float B2 = 2 * (y3 - y1);
float C2 = (distance1 * distance1 - distance3 * distance3) - (x1 * x1 - x3 * x3) - (y1 * y1 - y3 * y3);

float A3 = 2 * (x4 - x1);
float B3 = 2 * (y4 - y1);
float C3 = (distance1 * distance1 - distance4 * distance4) - (x1 * x1 - x4 * x4) - (y1 * y1 - y4 * y4);

estimatedPoint.x = (C1 * B2 - C2 * B1) / (A1 * B2 - A2 * B1);
estimatedPoint.y = (A1 * C2 - A2 * C1) / (A1 * B2 - A2 * B1);

estimatedPoint.x = (estimatedPoint.x + (C1 * B3 - C3 * B1) / (A1 * B3 - A3 * B1)) / 2;
estimatedPoint.y = (estimatedPoint.y + (A1 * C3 - A3 * C1) / (A1 * B3 - A3 * B1)) / 2;

return estimatedPoint;
}

```

Second, an example of Arduino files.

```

#include <WiFi.h>

const char* ssid = "MyESP32AP3";
const char* password = "1234567890";
const int channel = 1;
const int max_connection = 1;

void setup() {
  Serial.begin(115200);
  delay(10);

  // Start
  WiFi.softAP(ssid, password, channel, 0, max_connection);
  Serial.println("Access point started");
  Serial.print("SSID: ");
  Serial.println(ssid);
  Serial.print("IP address: ");
  Serial.println(WiFi.softAPIP());
}

void loop() {
}

```